CT-AI v2.0 • 2026/04/17 • ISTQB syllabus-based notes

CT-AI v2.0 — Seven-Chapter Study Companion

Independent English edition with every official subsection and concise key definitions.

These are independently written study summaries—not official translations or verbatim ISTQB text. The official syllabus remains the primary examination source.

Structure: 50 numbered subsections, seven concept maps, English key definitions and separate practical examples. **Page limit:** At most 5 pages per chapter.

Chapter 1 Chapter 2 Chapter 3 Chapter 4 Chapter 5 Chapter 6 Chapter 7

Chapter	Official chapter	Coverage
Chapter 1	Introduction to Artificial Intelligence	8 subsections
Chapter 2	Quality Characteristics for AI-Based Systems	3 subsections
Chapter 3	Machine Learning	12 subsections
Chapter 4	Testing AI-Based Systems	8 subsections
Chapter 5	Input Data Testing for Machine Learning Systems	7 subsections
Chapter 6	Model Testing for Machine Learning Systems	10 subsections
Chapter 7	Machine Learning Development Testing	2 subsections

Source: ISTQB Certified Tester AI Testing (CT-AI) Syllabus Version 2.0, General Assembly release April 17, 2026.

Method: Read the subsection bullets, revisit the concept map and definitions, then apply one measurable AI Interviewer test and record evidence.

CHAPTER 1 • 120 MINUTES • OFFICIAL SYLLABUS PP. 14–20

Introduction to Artificial Intelligence

مقدمة في الذكاء الاصطناعي

CHAPTER MIND MAP 1

1.1.1 AI-based system	**1.1.2** Narrow AI	**1.1.3** ML
1.1.4 GenAI	**1.1.5** CPU	**1.1.6** AI as a Service (AIaaS)
1.1.7 TensorFlow	**1.1.8** EU AI Act	

1.1 — Introduction to AI

1.1.1 AI-Based and Conventional Systems

English key terms: AI-based system, conventional system, data-driven, explainability

- Conventional software follows explicitly designed logic; machine-learned AI typically derives its decision behavior from data. Learned outputs may have uncertainty and cannot always be traced to a simple rule.
- Testing implications: verify AI functional performance on representative data, examine explainability and hidden bias, and continue to test ordinary software contracts such as permissions, validation and APIs.

1.1.2 Narrow AI, General AI, and Super AI

English key terms: Narrow AI, General AI / Strong AI, Super AI, technological singularity

- Narrow AI is designed for limited tasks and dominates real deployments. General AI is hypothetical broadly human-level intelligence across many tasks without task-by-task retraining.
- Super AI is a hypothetical self-improving form surpassing human intelligence; the potential transition is called a technological singularity. The syllabus states that no general AI system currently exists.

1.1.3 Different Types of AI Technologies

English key terms: ML, supervised / unsupervised / reinforcement learning, DL, CNN, RNN, transformer, NLP, agentic AI

- ML learns from examples; supervised learning uses labels, unsupervised learning finds structures and reinforcement learning uses rewards. Deep Learning employs multilayer neural networks; CNNs suit images and Transformers model sequences with attention.
- Other AI technologies include NLP, computer vision, fuzzy logic, search and rule-based expert systems. Agentic AI combines planning, reasoning and autonomous actions; tests must reflect technology-specific failure modes.

1.1.4 Generative AI

English key terms: GenAI, GAN, diffusion model, transformer, foundation model, multimodal

- Generative AI creates new text, images, audio, video or other content. GANs train competing networks, diffusion models iteratively denoise, and Transformers use attention; many products adapt pretrained foundation models.
- Evaluate output utility, factual and safety risks, deepfakes/misinformation, privacy, sustainability and employment impact. Multimodal systems need tests across more than one input/output modality.

1.1.5 Hardware for Machine Learning Systems

English key terms: CPU, GPU, AI-specific ASIC, edge, quantization, parallelism

- ML workloads benefit from large-data handling and parallel matrix operations. GPUs often outperform general CPUs in training; specialized AI hardware and edge accelerators may support low-power inference.
- Quantization uses fewer bits to reduce memory, cost, and energy, potentially changing numerical behavior. Test model quality, throughput, resources and compatibility on the actual target hardware.

1.1.6 Development and Hosting of AI Models

English key terms: AI as a Service (AIaaS), on-premises, cloud, edge, third-party model

- Options include building locally or using cloud/private infrastructure, third-party pretrained models, and AI as a Service. Choices trade control and data privacy against scale, setup cost, skills and rapid delivery.
- Test location-sensitive obligations, external dependency failures, privacy boundaries, performance and reproducibility; distinguish model training environment from inference/deployment environment.

1.1.7 Machine Learning Development Frameworks

English key terms: TensorFlow, PyTorch, scikit-learn, ML development framework

- Frameworks supply reusable algorithms, training, evaluation and model utilities. Selection depends on supported tasks, interfaces, hardware, skills, documentation, ecosystem, deployment and performance needs.
- Framework errors, misuse, updates and library dependencies may change behavior; validate APIs, evaluations, version compatibility and framework-specific limitations.

1.1.8 Regulations and Standards for AI

English key terms: EU AI Act, ISO/IEC standards, NIST AI RMF, risk-based regulation

- Applicable laws and standards guide responsible AI design, quality, transparency, privacy, safety and accountability. Their practical relevance depends on the system, jurisdiction and degree of risk.
- Derive test requirements from the actual regulatory context, including auditability, bias, human oversight, documentation and security. Do not treat an illustrative standard as universal legal advice.

Important English definitions

AI-based system
A system that uses artificial intelligence techniques for at least part of its behavior.

Narrow AI
AI intended to perform a limited set of tasks.

Generative AI (GenAI)
AI that produces new content such as text, images, or audio.

Agentic AI
AI involving autonomous agents that plan, reason, and act to meet goals.

Suggested THL AI Interviewer practice (applied example, not official syllabus text):
In THL, test that interview questions are selected from the saved question bank (exact assertion) while evaluating generated Observations with evidence-based criteria.

ISTQB® CT-AI v2.0 — version 2.0 GA, 17 Apr 2026, pp. 14–20. Original paraphrased study notes; official syllabus is the examination reference.

CHAPTER 2 • 45 MINUTES • OFFICIAL SYLLABUS PP. 21–25

Quality Characteristics for AI-Based Systems

خصائص جودة الأنظمة القائمة على الذكاء الاصطناعي

CHAPTER MIND MAP 2

2.1.1	2.1.2	2.2.1
ISO/IEC 25059	safety-related AI	AI acceptance criterion

2.1 — Quality Characteristics for AI-Based Systems

2.1.1 AI-Specific Quality Characteristics

English key terms: ISO/IEC 25059, ISO/IEC 25010, product quality, quality in use

- ISO/IEC 25059 extends ISO/IEC 25010: AI functional correctness allows defined error limits; functional adaptability is autonomous adaptation after deployment; AI robustness preserves functional correctness under adverse inputs and conditions.
- User controllability permits timely external control; intervenability targets timely harm prevention; transparency communicates relevant information. Societal and ethical risk mitigation addresses fairness, privacy, accountability, safety and human values (quality in use).

2.1.2 AI and Safety

English key terms: safety-related AI, non-determinism, incomplete specifications, harm mitigation

- Safety-critical systems are difficult to specify exhaustively when learning replaces explicit rules and behavior may evolve or be difficult to explain.
- Use operational limits, documented hazards, risk-driven verification, independent evidence, monitoring and timely intervention/rollback rather than relying solely on functional pass/fail tests.

2.2 — Acceptance Criteria for AI-Based Systems

2.2.1 Acceptance Criteria for AI-Based Systems

English key terms: AI acceptance criterion, measurable threshold, accuracy, recall, transparency, safe intervention

- Define measurable, risk-based acceptance criteria for both functional and AI-specific nonfunctional quality. Examples in the syllabus include 95% image-classification accuracy, 90% defect-prediction recall, intervention times, model-version endpoints and degraded-mode robustness.
- Thresholds are examples, not automatic requirements for your product. Specify metric, dataset, sample/uncertainty, environment, responsible approver and critical safety stop rules before deciding PASS or FAIL.

Important English definitions

AI functional correctness
The degree to which correct and incorrect AI outputs meet defined acceptable boundaries.

Functional adaptability
The ability to adapt autonomously to operational changes after deployment.

User controllability
Timely intervention in functioning by a human or other external agent.

Intervenability
Timely intervention by an operator to prevent harm or a hazard.

AI robustness
Ability to retain AI functional correctness in adverse conditions.

Transparency
Appropriate communication of system information to stakeholders.

Suggested THL AI Interviewer practice (applied example, not official syllabus text):
Create approved acceptance criteria for Observations: grounded claims, critical unsupported qualifications, reviewer intervention and fairness analysis. Proposed thresholds require Product/QA sign-off.

ISTQB® CT-AI v2.0 — version 2.0 GA, 17 Apr 2026, pp. 21–25. Original paraphrased study notes; official syllabus is the examination reference.

CHAPTER 3 • 375 MINUTES • OFFICIAL SYLLABUS PP. 26–37

Machine Learning

التعلم الآلي

CHAPTER MIND MAP 3

3.1.1 supervised learning	3.1.2 ML workflow	3.1.3 hands-on: supervised classifier
3.1.4 pretrained model	3.2.1 data preparation	3.2.2 hands-on: transform dataset for supervised classification
3.3.1 confusion matrix	3.3.2 hands-on: accuracy	3.3.3 hands-on: model comparison
3.4.1 perceptron	3.4.2 hands-on: AND gate	3.4.3 neuron coverage

3.1 — Introduction to Machine Learning

3.1.1 Different Forms of Machine Learning

English key terms: supervised learning, classification, regression, unsupervised learning, clustering, association, reinforcement learning

- Supervised learning maps labeled inputs to target outcomes (classification or numerical regression). Unsupervised learning discovers groups or associations without explicit target labels.
- Reinforcement learning learns a policy from actions, environment feedback and rewards. Each form requires a different source of expected behavior and corresponding testing approach.

3.1.2 Machine Learning Workflow

English key terms: ML workflow, training, validation / evaluation, tuning, independent test set

- The workflow includes objective definition, collecting/preparing data, model choice, training, evaluation and tuning, independent model testing, deployment and monitoring. Iteration is expected when acceptance is not met.
- Training fits parameters; evaluation/validation guides selection and hyperparameters; independent test data checks the accepted model against agreed metrics. Reusing the test set for tuning creates leakage and optimistic estimates.

3.1.3 Hands-on Exercise: Create a Machine Learning Model

English key terms: hands-on: supervised classifier, validation-versus-test accuracy

- Select a supervised classification algorithm, train it and test it with designated datasets. Record model configuration and measured accuracy.
- Compare validation accuracy used during tuning with independent test accuracy. Explain why the latter is evidence for generalization rather than another tuning signal.

3.1.4 Pretrained Models, Fine-Tuning, and Retrieval-Augmented Generation

English key terms: pretrained model, fine-tuning, retrieval-augmented generation (RAG)

- Pretrained models offer reusable learned capabilities; fine-tuning further trains model parameters using task-specific data. They save resources but may inherit model limitations and bias.
- RAG retrieves relevant external information and supplies it in the prompt/context at inference time. Evaluate retrieval quality, relevance and the model's grounding in returned sources; do not confuse RAG with retraining.

3.2 — Data for Machine Learning

3.2.1 Activities in Data Preparation

English key terms: data preparation, feature engineering, class balancing, train/validation/test split, data leakage

- Prepare data by cleaning invalid and duplicate values, dealing with missing entries, selecting/engineering features, transforming/scaling where appropriate, labeling and addressing imbalanced classes.
- Choose statistically appropriate split or cross-validation strategies and keep the final holdout isolated. Representativeness and dataset size affect the reliability of a trained model and later performance measurements.

3.2.2 Hands-on Exercise: Data Preparation in Support of the Creation of a Machine Learning Model

English key terms: hands-on: transform dataset for supervised classification

- Given a provided dataset and development tools, perform relevant cleaning, encoding or feature preparation steps and make it fit for a supervised classification model.
- Preserve data-handling decisions and a valid train/test separation for subsequent model-building exercises.

3.3 — ML Functional Performance Metrics for Classification

3.3.1 Calculation of Machine Learning Functional Performance Metrics

English key terms: confusion matrix, TP/FP/FN/TN, accuracy, precision, recall, F1-score

- Confusion matrix identifies TP, FP, FN and TN. Accuracy=(TP+TN)/N; precision=TP/(TP+FP); recall/sensitivity=TP/(TP+FN); F1=2×precision×recall/(precision+recall). Note the actual/predicted axis orientation.
- Choose metrics according to error cost and class distribution. High overall accuracy can conceal failures on rare classes; review per-class results, false alarms and missed detections.

3.3.2 Hands-on Exercise: Evaluate a Machine Learning Model using Selected ML Functional Performance Metrics

English key terms: hands-on: accuracy, precision, recall, F1

- Evaluate the classifier developed earlier; calculate and display accuracy, precision, recall and F1 using an ML framework or verified library functions.
- Interpret which mistake types dominate the result and explain whether the selected metrics suit the application’s risk.

3.3.3 Hands-on Exercise: Show the Impact of Different Machine Learning Models and Dataset Combinations

English key terms: hands-on: model comparison, dataset combination, training duration

- Use alternative classifiers and different dataset combinations to observe changes in model generation, training time and final model behavior.
- Compare final ML functional performance metrics under controlled conditions; do not assume the most complex algorithm is the best.

3.4 — Neural Networks

3.4.1 Structure and Working of a Deep Neural Network

English key terms: perceptron, input, hidden, output layers, weights, bias, activation, epoch

- Deep neural networks contain input, one or more hidden and output layers; weighted connections, biases and activation functions transform the signal. Learning updates weights to reduce error over epochs.
- A perceptron supports linearly separable binary classification; deep multilayer structures model more complex patterns but create additional testability and interpretation challenges.

3.4.2 Hands-on Exercise: Experience the Implementation of a Perceptron

English key terms: hands-on: AND gate, weight and bias updates

- Simulate a perceptron that learns the AND Boolean function, using a spreadsheet, visual tool or simple program.
- Observe how weight and bias updates across epochs reduce the classification error to zero for this linearly separable task.

3.4.3 Coverage Measures for Neural Networks

English key terms: neuron coverage, neuron boundary coverage, k-multisection neuron coverage

- Neuron coverage measures whether selected neuron activation criteria are exercised. Neuron boundary coverage examines crossing defined activation boundaries; k-multisection coverage partitions a neuron’s activation range.
- Use internal coverage to highlight unexplored input behaviors but avoid equating coverage percentages with correct predictions, sufficient test-data representativeness or safety.

Important English definitions

Supervised learning
Learning a mapping from labeled inputs to expected target values.

Validation dataset
Data used to evaluate model choices and guide tuning.

Test dataset
Data reserved for independent final performance assessment.

Precision
The proportion of predicted positives that are actually positive.

Recall
The proportion of actual positives that are correctly identified.

F1-score
The harmonic mean of precision and recall.

RAG
Retrieval-Augmented Generation: conditioning generation on retrieved external content.

Applied THL exercise
Use synthetic candidate_evaluation.csv and Python to calculate TP, FP, FN, TN, precision, recall and F1 against independent reviewer labels.

ISTQB® CT-AI v2.0 — version 2.0 GA, 17 Apr 2026, pp. 26–37. Original paraphrased study notes; official syllabus is the examination reference.

CHAPTER 4 • 195 MINUTES • OFFICIAL SYLLABUS PP. 38–45

Testing AI-Based Systems

اختبار الأنظمة المعتمدة على الذكاء الاصطناعي

CHAPTER MIND MAP 4

4.1.1 locked AI	4.1.2 statistical testing	4.1.3 test oracle problem
4.2.1 GenAI black-box testing	4.2.2 red teaming (RT)	4.2.3 hands-on: exploratory testing
4.3.1 ML input data testing	4.3.2 risk-based testing	

4.1 — Introduction to Testing AI-Based Systems

4.1.1 Locked and Adaptive AI-Based Systems

English key terms: locked AI, adaptive AI, online learning, operational change

- A locked AI system holds its learned behavior fixed after deployment unless a controlled update is applied. Adaptive AI alters itself using operational inputs and may change while tests execute.
- For adaptive systems, automated regression around stable core functions, version control, change monitoring and operational testing are important; not every future behavior can be predicted at design time.

4.1.2 Rationale for a Statistical Approach to Testing AI-Based Systems

English key terms: statistical testing, error rate, representative sample, confidence

- Probabilistic predictions make a single correct output insufficient evidence. Define acceptance as performance across a representative sample, with an explicit error rate and uncertainty.
- Use repeated tests, statistical aggregation and adequate sample sizes where needed; interpret outcomes by operational scenario and severity, not only average performance.

4.1.3 Test Oracles for AI-Based Systems

English key terms: test oracle problem, human expert, ground truth, pseudo-oracle, metamorphic relation

- The test-oracle problem occurs when a precise expected outcome is absent, subjective, too costly or difficult to compute. AI requirements may also be incomplete and evolve during exploratory development.
- Combine independent expert judgments, reliable ground truth, statistical baselines, reference implementations and metamorphic relations. Assess the limitations and possible shared biases of each oracle.

4.2 — Testing Generative AI and Large Language Models

4.2.1 Testing Generative AI

English key terms: GenAI black-box testing, input explosion, benchmarks, LLM-as-judge

- Test prompts, context, system instructions, generation settings, multimodal data and APIs; the combinatorial input space creates an input-explosion problem. Assess clarity, domain rules, usefulness and content quality.
- Use benchmarks, human review or a second model as a cautious assessor; a second model may share errors. Include latency, memory, CPU/GPU usage, throughput and operational cost in nonfunctional testing.

4.2.2 Red Teaming

English key terms: red teaming (RT), adversarial prompt, indirect prompt injection, safety/privacy attack

- Red teaming deliberately probes an AI system for undesirable or harmful outputs and misuse capabilities. It is often black-box and adaptive: investigators refine prompts in response to findings.
- Target security, privacy, fairness, disinformation and unsafe behavior; include prompt injection and malicious content from external retrieved documents. Use results to improve safeguards and retest.

4.2.3 Hands-on Exercise: Exploratory Testing of a Large Language Model

English key terms: hands-on: exploratory testing, 2-value/3-value boundary value analysis

- The official exercise runs a time-boxed exploratory session on an LLM asked to generate test cases for 2-value and 3-value boundary value analysis.
- Use a session sheet and debrief to assess both correctness and completeness of generated boundary test cases, including missing and invalid partitions.

4.3 — Test Levels and Machine Learning Systems

4.3.1 Test Levels for Machine Learning Systems

English key terms: ML input data testing, ML model testing, component, integration, system, acceptance

- ML introduces special levels for input-data testing and trained-model testing in addition to conventional component, component-integration, system and acceptance tests.
- Trace test evidence through data preparation, model behavior, API integration and end-user workflows; a model that meets its metrics may still be embedded in a defective system.

4.3.2 Risk-Based Testing of Machine Learning Systems

English key terms: risk-based testing, likelihood, impact, AI harm

- Prioritize test design and depth using the likelihood and impact of AI failures, including fairness, safety, security, performance and inaccurate predictions.
- Give high-risk scenarios representative datasets, stronger oracles and stricter criteria. Revisit risk assessment when the model, deployment context or operational distribution changes.

Important English definitions

Locked AI
An AI system whose learned behavior is fixed after deployment until an explicit update.

Adaptive AI
An AI system that changes its behavior in operation based on newly available information.

Test oracle
A source used to determine whether an observed test result is correct.

Oracle problem
The difficulty of deciding the expected correct result for an AI test.

Red teaming
Systematic, adversarial probing for harmful outputs or weak safeguards.

Suggested THL AI Interviewer practice (applied example, not official syllabus text):
Compare a generated interview report with recording/transcript evidence, repeat the same answer multiple times and probe unsafe prompt instructions.

ISTQB® CT-AI v2.0 — version 2.0 GA, 17 Apr 2026, pp. 38–45. Original paraphrased study notes; official syllabus is the examination reference.

CHAPTER 5 • 180 MINUTES • OFFICIAL SYLLABUS PP. 46–53

Input Data Testing for Machine Learning Systems

اختبار بيانات الإدخال لأنظمة التعلم الآلي

CHAPTER MIND MAP 5

- 5.1.1 data poisoning
- 5.1.2 data bias
- 5.1.3 ETL
- 5.1.4 representativeness
- 5.1.5 single-value constraints
- 5.1.6 label correctness
- 5.1.7 hands-on: missing records

5.1 — Input Data Testing for Machine Learning Systems

5.1.1 Input Data Risks and Mitigations

English key terms: data poisoning, provenance, data quality, pipeline defect

- Training and inference data can be damaged by poisoning, unreliable provenance, missing or inconsistent values, biased coverage, wrong constraints, incorrect labels or pipeline failure.
- Match each risk to targeted controls: review source and labels, analyze distributions, enforce schema/constraints, test the pipeline and check security and performance.

5.1.2 Testing for Bias

English key terms: data bias, algorithmic bias, disparate impact, subgroup testing

- Bias can enter from historical or unrepresentative training data, labeling, data collection, algorithm design or decision thresholds. Review datasets, procedures and documentation for systematic skew.
- Use statistical distribution checks, representative evaluation inputs and subgroup outcome comparisons to identify differential model behavior. Distinguish observed disparity from its underlying cause.

5.1.3 Data Pipeline Testing

English key terms: ETL, schema test, transformation test, drift monitor, back-to-back

- Test acquisition, transformations, cleaning, feature computation, storage and transfer; verify schema, ranges, completeness, duplication, security, throughput and resulting data quality.
- At production scale, monitor pipelines dynamically and compare versions with back-to-back or A/B methods to catch regressions and hidden changes to labels or features.

5.1.4 Testing for Data Representativeness

English key terms: representativeness, exploratory data analysis (EDA), distribution, KS / Chi-square tests

- Representativeness means the dataset reflects the target use cases and operational input distributions. Inspect descriptive statistics, subgroup balance, relationships, anomalies and coverage gaps.
- Use EDA, statistical comparisons such as Chi-square or Kolmogorov–Smirnov where appropriate, and targeted test data for underrepresented classes. Test data also needs operational validity and independence.

5.1.5 Dataset Constraint Testing

English key terms: single-value constraints, multi-value constraints, comparison constraints

- Check permissible values, types, nullability, uniqueness, ranges and formats, plus relationships across attributes. Comparison constraints can enforce greater-than or correlation expectations.
- Automate constraint checks in the data pipeline so anomalies are detected at scale and recorded for data scientists and operations; constraints complement but do not replace representativeness testing.

5.1.6 Label Correctness Testing

English key terms: label correctness, multiple annotation, inter-annotator agreement (IAA), Cohen’s kappa

- Wrong labels degrade supervised ML generalization. Validate by domain-expert review, independent multiple annotations, rule checks, label-distribution comparison, loss analysis and low-confidence cases.
- Calculate inter-annotator agreement, e.g. Cohen’s kappa, and adjudicate disagreement. Focus extra review on ambiguous or high-risk examples rather than trusting model predictions as truth.

5.1.7 Hands-on Exercise: Input Data Testing

English key terms: hands-on: missing records, duplicates, outliers

- The official practical task checks a provided dataset—typically structured/tabular—for missing fields, duplicate records and outliers.
- Report detected defects, document validation rules and make sure pipeline corrections do not silently remove valid rare cases.

Important English definitions

Data provenance
Evidence of a dataset's origin and processing history.

Representativeness
Degree to which a dataset reflects its target operational population and scenarios.

Dataset constraint
A rule restricting valid values or relationships in data.

Label correctness
Accuracy and consistency of ground-truth annotations.

Inter-annotator agreement (IAA)
Agreement between independent people assigning labels to the same examples.

Suggested THL AI Interviewer practice (applied example, not official syllabus text):
Check missing transcripts, duplicated answer rows, Arabic/English balance, source permissions and independent expert labels before running an evaluation.

ISTQB® CT-AI v2.0 — version 2.0 GA, 17 Apr 2026, pp. 46–53. Original paraphrased study notes; official syllabus is the examination reference.

CHAPTER 6 • 225 MINUTES • OFFICIAL SYLLABUS PP. 54–62

Model Testing for Machine Learning Systems

اختبار نماذج التعلم الآلي

CHAPTER MIND MAP 6

- **6.1.1** model risk
- **6.1.2** model card
- **6.1.3** confidence level (CL)
- **6.1.4** adversarial example
- **6.1.5** metamorphic relation (MR)
- **6.1.6** hands-on: derive MRs
- **6.1.7** data drift
- **6.1.8** overfitting
- **6.1.9** A/B testing
- **6.1.10** pseudo-oracle

6.1 — Model Testing for Machine Learning Systems

6.1.1 Machine Learning Model Risks and Mitigations

English key terms: model risk, bias, overfitting, adversarial vulnerability, poor performance

- Model risks include wrong predictions, bias, under/overfitting, adversarial susceptibility, lack of robustness, slow responses, poor documentation and regression after updates.
- Combine functional performance testing, bias/adversarial tests, fuzzing, performance checks, documentation review, drift monitoring and controlled model comparisons.

6.1.2 Machine Learning Model Documentation and Review

English key terms: model card, design/usage limits, model documentation review

- Model documentation should identify purpose, author/version/license, design assumptions, algorithm, intended and prohibited uses, training/evaluation data, interfaces and measured performance.
- Review constraints, environmental requirements, fairness/safety limitations and maintenance history so testers can choose valid inputs, trustworthy oracles and meaningful release criteria.

6.1.3 ML Functional Performance Testing of Probabilistic Machine Learning Systems

English key terms: confidence level (CL), margin of error (MoE), sample size, holdout performance

- Probabilistic performance needs sufficiently large independent test datasets. Specify expected metric, confidence level (CL), margin of error (MoE) and sample size; smaller error margins need larger samples.
- The syllabus illustrates checking 98% accuracy with ±4% MoE at 95% CL using 601 cases under a conservative proportion-based calculation. Such figures are examples, not universal release gates.

6.1.4 Adversarial Testing of Machine Learning Systems

English key terms: adversarial example, black-box/white-box attack, transferability

- Adversarial examples are deliberately modified inputs that cause incorrect model behavior, sometimes through small changes that seem harmless to people.
- Attack strategies may use model internals (white-box) or output-only queries/proxy models (black-box). Probe realistic threat models, robustness and recovery rather than treating one attack success as a complete assessment.

6.1.5 Metamorphic Testing

English key terms: metamorphic relation (MR), source test, follow-up test

- When an exact expected output is hard to know, define a Metamorphic Relation (MR) between inputs and corresponding outputs. Derive follow-up tests from a source case and a valid input transformation.
- Some MRs require the same output (e.g., an irrelevant name change leaves competency unchanged); others require a predictable change. Invalid or unsafe MRs can create misleading failures, so validate the relation.

6.1.6 Hands-on Exercise: Apply Metamorphic Testing

English key terms: hands-on: derive MRs, generate source/follow-up tests, execute

- Derive multiple MRs for an MLS, including both same-output and changed-output expectations. Generate source cases, then derive transformed follow-up cases.
- Execute follow-ups and compare relationships rather than individual outputs; understand the limitation when the original source case has not independently passed.

6.1.7 Drift Testing

English key terms: data drift, concept drift, monitoring, operational distribution

- Drift means operational inputs or the statistical relationship between inputs and target outcomes changes relative to training assumptions. Accuracy may degrade even without a software change.
- Compare current input and performance distributions to baselines, identify change patterns and trigger investigation or retraining where appropriate; do not treat all shifts as defects.

6.1.8 Testing for Overfitting and Underfitting

English key terms: overfitting, underfitting, learning curve, validation gap

- Overfitting means learning training-specific noise and generalizing poorly, often visible as a large difference between training and unseen data performance. Underfitting means insufficiently capturing the relationship and poor performance even on training data.
- Inspect learning curves, training/validation/test performance and relevant error metrics; consider model capacity, data quality, regularization and available training evidence.

6.1.9 A/B Testing

English key terms: A/B testing, statistical comparison, control vs variant

- A/B testing exposes comparable groups or inputs to variants A and B to decide which is better using agreed ML performance measures and statistical evidence.
- It helps compare updates and adaptive changes but does not itself generate cases or prove absolute correctness; account for sample size, randomization, confounders and the preapproved decision rule.

6.1.10 Back-to-Back Testing

English key terms: pseudo-oracle, back-to-back comparison, independent implementation

- Run the same inputs against the system under test and a reference (pseudo-oracle) version; compare outputs to detect deviations, especially after migration or model changes.
- Prefer independently implemented references to avoid shared defects. Unlike A/B, back-to-back mainly detects differences/defects; equal outputs can still both be wrong.

Important English definitions

Adversarial example
An intentionally manipulated input that induces incorrect model behavior.

Metamorphic relation (MR)
An expected relation between outputs when an input undergoes a defined transformation.

Data drift
A change in the statistical distribution of incoming data.

Concept drift
A change in the relationship between inputs and target outcomes.

Margin of error (MoE)
A measure of uncertainty around a statistical estimate.

Confidence level (CL)
A selected long-run coverage level for an estimation procedure.

Back-to-back testing
Comparison with a reference version or pseudo-oracle using the same test inputs.

Suggested THL AI Interviewer practice (applied example, not official syllabus text):
Apply paraphrase-based metamorphic relations, compare model versions on fixed goldens, evaluate fairness and flag unsupported hiring judgments.

ISTQB® CT-AI v2.0 — version 2.0 GA, 17 Apr 2026, pp. 54–62. Original paraphrased study notes; official syllabus is the examination reference.

CHAPTER 7 • 30 MINUTES • OFFICIAL SYLLABUS PP. 63–66

Machine Learning Development Testing

اختبار تطوير أنظمة التعلم الآلي

CHAPTER MIND MAP 7

- **7.1.1** ML framework API
- **7.1.2** installability

7.1 — Machine Learning Development Testing

7.1.1 Machine Learning Development Risks and Mitigations

English key terms: ML framework API, hyperparameters, evaluation implementation, build security

- Development risks include framework/API misuse, unsuitable algorithms/hyperparameters, library defects, corrupt data allocation, installation/build failures, insecure dependencies, poor documentation and misleading stochastic evaluation.
- Mitigate by API/smoke/security/performance tests; review framework and selection choices, check split allocation and independently cross-check reported ML performance and model comparisons.

7.1.2 Machine Learning System Deployment Testing

English key terms: installability, rollback, canary, shadow, model conversion, cross-device, API testing

- Deployment checks include install/configure/uninstall where supported, dependency and target-environment compatibility, API behavior, model conversion quality, resource efficiency and correctness across devices.
- Rollback tests confirm recovery before release; canary exposes a small production share while monitoring; shadow executes a new model alongside the incumbent without using its result for live decisions. Select techniques according to deployment risks.

Important English definitions

Canary testing
Limited rollout to a subset of production traffic with monitored metrics.

Shadow testing
Parallel execution of a new model without using its outputs for live decisions.

Rollback testing
Verification that an update can be reverted safely.

Model conversion testing
Verification that conversion preserves acceptable behavior and efficiency.

Suggested THL AI Interviewer practice (applied example, not official syllabus text):
Test model/report version compatibility and API permissions; if shadow or canary rollout exists, compare results without exposing unapproved output to real hiring decisions.

ISTQB® CT-AI v2.0 — version 2.0 GA, 17 Apr 2026, pp. 63–66. Original paraphrased study notes; official syllabus is the examination reference.